# One Interval, Two Threshold Claims

AP Statistics · Unit 3 · Topic 3.4 · B: 2026-12-01 · E: 2027-01-06 · TEACHER KEY

## CED TETHER

- Enduring Understanding: UNC-4 An interval of values should be used to estimate parameters, in order to account for uncertainty.
- Skills:
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **Skill 4.D** — Justify a claim based on a confidence interval.
- **Skill 4.A** — Make an appropriate claim or draw an appropriate conclusion.
- **LO UNC-4.F** — Interpret a confidence interval for a population proportion. [Skill 4.B]
- **EK UNC-4.F.1** — A confidence interval for a population proportion either contains the population proportion or it does not, because each interval is based on random sample data, which varies from sample to sample.
- **EK UNC-4.F.2** — We are C% confident that the confidence interval for a population proportion captures the population proportion.
- **EK UNC-4.F.3** — In repeated random sampling with the same sample size, approximately C% of confidence intervals created will capture the population proportion.
- **EK UNC-4.F.4** — Interpreting a confidence interval for a one-sample proportion should include a reference to the sample taken and details about the population it represents.
- **LO UNC-4.G** — Justify a claim based on a confidence interval for a population proportion. [Skill 4.D]
- **EK UNC-4.G.1** — A confidence interval for a population proportion provides an interval of values that may provide sufficient evidence to support a particular claim in context.
- **LO UNC-4.H** — Identify the relationships between sample size, width of a confidence interval, confidence level, and margin of error for a population proportion. [Skill 4.A]
- **EK UNC-4.H.1** — When all other things remain the same, the width of the confidence interval for a population proportion tends to decrease as the sample size increases. For a population proportion, the width of the interval is proportional to $1/\sqrt{n}$.
- **EK UNC-4.H.2** — For a given sample, the width of the confidence interval for a population proportion increases as the confidence level increases.
- **EK UNC-4.H.3** — The width of a confidence interval for a population proportion is exactly twice the margin of error.

**Essential question:** How can one confidence interval support a modest lower-bound claim but not a stronger one?

**DOK-3 skill:** 4.D · **FRQ pattern:** `one-interval-two-threshold-claims-bound-the-conclusion`

**DOK rationale:** Part (c) coordinates interval construction, contextual interpretation, and two endpoint comparisons to adjudicate competing conclusions, distinguish plausibility from support, and bound what can be reported.

**Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK:** 1 → 3 **Minutes:** 45

**Teacher does**

- Board slide up at the bell. Ask for a point-estimate first take.
- Begin the video at minute 5; return at minute 33 to compare the lower endpoint with both thresh-

olds.

- Collect at the door. Score part (c) only, E/P/I.

**Students do**

- Choose which threshold claims have support and cite one number.
- Complete the follow-along, finish (a)–(c), revise, and turn in the sheet.

**Questions to ask**

- What uncertainty disappears if you report only 0.68?
- Which endpoint should you compare with a lower-bound claim?
- What does 0.65 inside the interval establish—and not establish?

**Adult role**

Solo teacher. Circulate 5 pairs. Require both endpoint comparisons and a bounded conclusion.

**[WARN] WATCH FOR**

- Treating the sample proportion as the population proportion.
- Calling an interior threshold a proved lower bound.
- Assigning 95 percent probability to the fixed parameter.

## SCORING PART (c) — E / P / I

**Expected elements**

- **judgment-1** (required) — Supports the more-than-60-percent claim using lower endpoint about 0.627 above 0.60.
- **judgment-2** (required) — Withholds the at-least-65-percent claim because the lower endpoint is below 0.65, while 0.65 remains plausible.
- **judgment-3** (required) — Supports Mira and explains that Kai's point estimate alone ignores sampling uncertainty.

| | |
|---|---|
| **E** | Meets all three checks with correct contextual reasoning; equivalent wording is acceptable. |
| **P** | Shows at least one substantive correct check, but has an omission or error that prevents E. |
| **I** | Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient. |

## [STAR] TODAY'S PROBLEM — annotated

Suppose a statewide activities association has 10,000 member students and is considering requiring reusable food containers at its tournaments. In a simple random sample of 300 member students, 204 favor the requirement. A sponsor claims that at least 65% of all member students favor it; an event organizer makes the more modest claim that more than 60% favor it.

**Kai:** "The sample has $204/300 = 68\%$ in favor, above both thresholds, so it supports both claims."

**Mira:** "The point estimate alone cannot settle both claims. We should construct a confidence interval and compare all of its plausible values with each threshold."

**Sample counts**

| Group | N | n | Favor | Other |
|---|---|---|---|---|
| **Members** | 10,000 | 300 | 204 | 96 |

### FIRST TAKE (5 min)

Do the data appear to support both claims, only one, or neither? Cite one number and commit before calculating the interval.

**Key:** Not graded. The interval must separate plausible from justified.

*Rules callout "LET THE WHOLE INTERVAL SET THE BOUND (keep on the page)" is printed on the student sheet.*

**DOK 1** **(a)** Define the population proportion $p$ and calculate the sample proportion $\hat{p}$.

**Key:** Let $p$ be the proportion of all 10,000 association member students who favor requiring reusable food containers at tournaments. The sample proportion is $\hat{p} = 204/300 = 0.68$.

**DOK 2** **(b)** Verify the conditions, construct a 95 percent confidence interval for $p$, and interpret the interval in the association's context.

**Key:** The association used an SRS; $300 \leq 0.10(10{,}000) = 1{,}000$; and there are 204 successes and 96 failures, both at least 10. Thus $SE = \sqrt{0.68(0.32)/300} = 0.0269320$ and $ME = 1.96(SE) = 0.0527867$. The 95 percent interval is $(0.6272, 0.7328)$, or about $(0.627, 0.733)$. We are 95 percent confident that this interval captures the proportion of all association member students who favor the reusable-container requirement.

**DOK 3** **(c)** In 3–4 sentences, decide which threshold claim the interval supports and whose reasoning is better. Compare the lower endpoint with 0.60 and 0.65. Explain why a plausible 0.65 is different from establishing at least 0.65.

**Key:** Mira is warranted. The entire interval $(0.627, 0.733)$ exceeds 0.60, supporting the organizer's claim. Its lower endpoint is below 0.65, so the interval does not establish the sponsor's lower bound. Because 0.65 is inside, it remains plausible rather than disproved. Kai treats 0.68 as exact, hiding the margin of error and falsely suggesting that both population bounds are established.

### EXIT

One thing the video changed about my first take: ______________________________